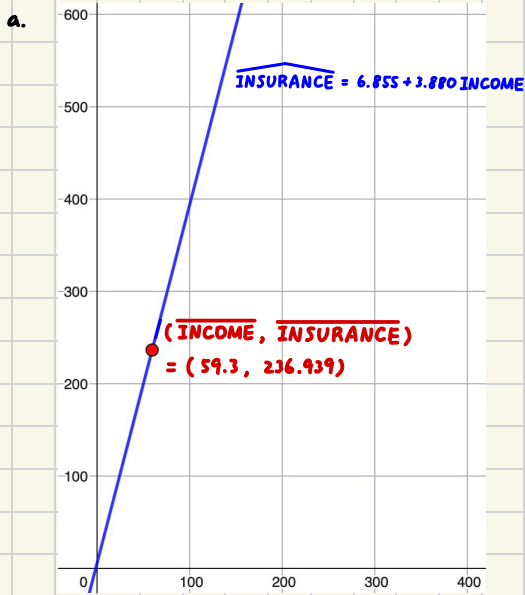


3.18 A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let *INCOME* be household income (thousands of dollars) and *INSURANCE* the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

$$\widehat{INSURANCE} = 6.855 + 3.880INCOME$$

(se) (7.383) (0.112)

- a. Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of $INCOME = 59.3$. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.
- b. How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.
- c. Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is -0.746 .
- d. One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $INSURANCE = \beta_1 + \beta_2 INCOME + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.
- e. Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?



b.

$$\widehat{INSURANCE} = 6.855 + 3.880 INCOME$$

(thousands of dollars) (thousands of dollars)

$$\Rightarrow INCOME + \$1,000, INSURANCE + \$3,880$$

$$d.f. = N - 2 = 18$$

95% C.I. of $\hat{\beta}_2$:

$$\begin{aligned} \hat{\beta}_2 \pm t_{0.025, 18} \cdot se(\hat{\beta}_2) \\ = 3.880 \pm 2.101 \cdot 0.112 \\ \approx [3.645, 4.115] \end{aligned}$$

point estimate: \$3,880

95% interval estimate: [\$3,645, \$4,115]

c.

Let $\begin{cases} \hat{y} = \widehat{INSURANCE} \\ x = INCOME \end{cases}$

$$\Rightarrow \hat{y} = \hat{\beta}_1 + \hat{\beta}_2 x; \hat{\beta}_1 = 6.855, \hat{\beta}_2 = 3.880$$

Substitute $\Rightarrow 6.855 + 3.880 \cdot 100 = 394.855$

$$\begin{aligned} Var(\hat{y}) &= Var(\hat{\beta}_1 + \hat{\beta}_2 x) \\ &= Var(\hat{\beta}_1) + x^2 Var(\hat{\beta}_2) + 2x Cov(\hat{\beta}_1, \hat{\beta}_2) \end{aligned}$$

Substitute $\Rightarrow (7.383)^2 + 100^2 \cdot (0.112)^2 + 2 \cdot 100 \cdot (-0.746)$

$$= 30.748689$$

99% C.I. of $\hat{y}$:

$$\begin{aligned} \hat{y} \pm t_{0.005, 18} \cdot se(\hat{y}) \\ = 394.855 \pm 2.878 \cdot \sqrt{30.748689} \\ \approx [378.896, 410.814] \end{aligned}$$

point estimate: \$394,855

interval estimate: [\$378,896, \$410,814]

d.

$$H_0: \beta_2 = 5$$

①

$$H_A: \beta_2 \neq 5$$

②

$$\alpha = 0.05$$

③

$$\text{test statistic: } t = \frac{\hat{\beta}_2 - 5}{\text{se}(\hat{\beta}_2)} \stackrel{H_0}{\sim} t(20-2)$$

④

$$\text{rejection region (RR)} = \{ |t| \geq t_{0.025, 18} = 2.101 \}$$

⑤

$$t_0 = \frac{3.880 - 5}{0.112} = -10$$

⑥

$$t_0 = -10 \in \text{RR}, \text{ reject } H_0$$

The data shows that for every \$1,000 increase in income the average amount of life insurance held will increase significantly different from \$5,000 at a 95% confidence level.

e.

①

$$H_0: \beta_2 = 1$$

②

$$H_A: \beta_2 > 1$$

③

$$\text{test statistic: } t = \frac{\hat{\beta}_2 - 1}{\text{se}(\hat{\beta}_2)} \stackrel{H_0}{\sim} t(20-2)$$

④

$$\text{RR} = \{ t \geq t_{0.01, 18} = 2.552 \}$$

⑤

$$t_0 = \frac{3.880 - 1}{0.112} \approx 25.714$$

⑥

$$t_0 = 25.714 \in \text{RR}, \text{ reject } H_0$$

We have sufficient sample evidence to indicate that as income increases the amount of life insurance increases significantly larger than the same amount at the 99% confidence level.